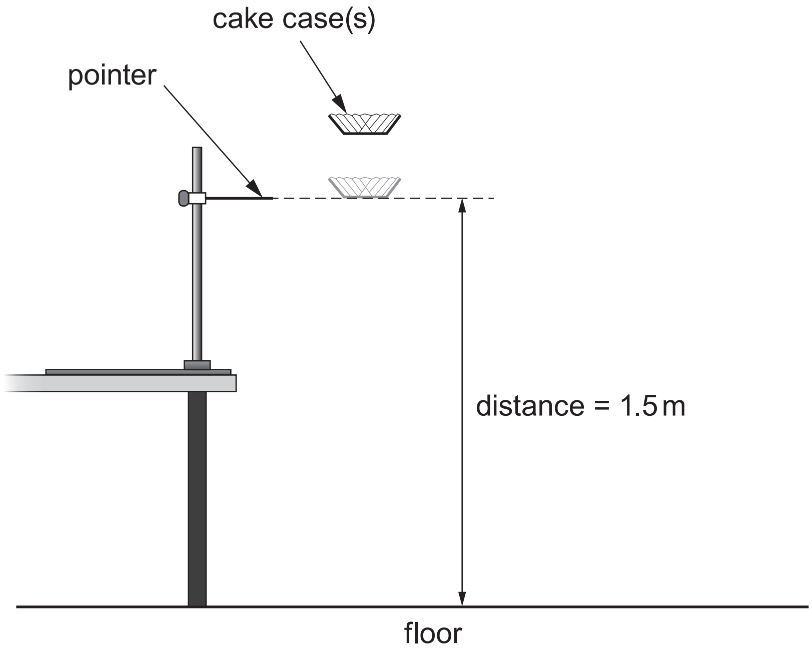
5. Students investigate the terminal speed of falling paper cake cases.
The apparatus they use is shown below.



Their results are given in the table.

Number of cake cases	Time to fall (s)				
	Trial 1	Trial 2	Trial 3	Mean	Terminal speed (m/s)
1	0.85	0.94	0.91	0.90	1.7
2	0.68	0.62	0.65	0.65	2.3
3	0.58	0.62	0.57	0.59	2.5
4	0.52	0.26	0.54	0.44	3.4
6	0.44	0.48	0.46	0.46	3.3

Examiner only

Use the results in the table to answer the following questions.

(a) (i) **Circle the anomalous result**, in the table, for the time to fall. [1]

(ii) Tom says that the mean time for 4 cake cases to fall is wrong.
Explain whether Tom is correct. [2]
Space for calculation.

....

....

(b) Tom thinks that as the number of cake cases doubles the terminal speed doubles.
Explain whether you agree with Tom. [2]

....

....

....

(c) (i) For 6 cases the times taken to fall are 0.44 s, 0.48 s and 0.46 s.

Use the equation:

$$\text{uncertainty} = \frac{\text{maximum value} - \text{minimum value}}{2}$$

to calculate the uncertainty in the measurements of time for 6 cake cases to fall. [2]

uncertainty = ... s

(ii) State **one** random error that is a cause of uncertainty when doing this experiment. [1]

....

8